

2. Domain Background

To understand a Payment Gateway system design, it is important to first understand the ecosystem in which it operates. The payment processing domain involves multiple financial entities, strict regulations, and secure communication channels.

What is a Payment Gateway (e.g., Stripe)?

A payment gateway like **Stripe** is **not a bank**. Instead, it acts as a **financial intermediary** that facilitates the movement of money between:

- The **customer's bank account/card**
- The **merchant's bank account**

It securely handles payment data, communicates with financial networks, and ensures that transactions are authorized and eventually settled.

In simple terms:

Stripe = Secure bridge between customers, banks, and merchants.

Why Not Build Your Own Payment System?

In theory, any business could try to build its own payment processing system. However, in practice, it is extremely difficult due to regulatory, technical, and financial constraints.

Key Challenges of Building a Payment System

1. Licensing and Compliance

To process card payments directly, a company must obtain multiple financial licenses, which may include:

- Payment institution licenses
- Banking partnerships
- PCI-DSS compliance certification (for handling card data securely)

These requirements vary by country and are expensive and time-consuming to obtain.

2. Security and Compliance Standards

Handling sensitive financial data (like card numbers) requires strict compliance:

- Encryption at rest and in transit

- Secure storage of card data (or avoiding storage via tokenization)
- Fraud prevention mechanisms
- Regular security audits

Failure to comply can result in:

- Heavy regulatory fines
- Loss of license
- Legal liability

3. Integration Complexity

A payment system must integrate with:

- Card networks (e.g., Visa, Mastercard)
- Issuing banks (customer's bank)
- Acquiring banks (merchant's bank)
- Fraud detection systems

Each integration has its own protocols, latency requirements, and failure modes.

4. Operational Risk

Payments systems are highly sensitive:

- Even small downtime leads to financial loss
- Incorrect transactions can result in chargebacks and disputes
- Systems must be highly available and consistent

Why Companies Use Stripe Instead

Instead of building and maintaining all of this infrastructure, businesses use payment gateways like Stripe because they provide:

- Pre-built compliance and licensing
- Secure handling of card data
- Ready-to-use bank and card network integrations
- Fraud detection systems
- Reliable settlement and reconciliation processes

This allows businesses to:

Focus on their core product instead of financial infrastructure.